THE FUNCTOR OF POINTS WHERE A MODULE IS FLAT IN DIMENSIONS AT LEAST N IS UNCHANGED BY ETALE LOCALISATION ON THE SOURCE WHEN THE ETALE IMAGE CONTAINS THE RELATIVE ASSOCIATED POINTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 05UD (sources/stacks-flat/pinned-original-source.tex line 7728).

Allowed standard prerequisites (editorial).

- The flatness-in-dimension-n situation, the functor F_n and its defining equation. Same chapter, flat.tex, Situation labelled situation-flat-dimension-n (source line 6311) and Equation labelled equation-flat-dimension-n (source line 6325), invoked in the statement. Fix the objects f : X -> S, the module F and the functor F_n whose subfunctor relation and agreement are the content of the lemma.
- Flatness permanence along an etale morphism and base change of relative associated points. Morphisms, Lemma labelled morphisms-lemma-flat-permanence (Stacks tag 02JZ) at proof lines 7746 and 7770; Divisors, Lemma labelled divisors-lemma-base-change-relative-assassin (Stacks tag 05DC) and Remark labelled divisors-remark-base-change-relative-assassin (Stacks tag 05KL) at proof lines 7763-7764. Supply the identification of the non-flatness loci in the fibre and the description of the fibre associated points as base changes of relative associated points.
- Flatness is checked on the associated points of the source. Same chapter, Theorem labelled theorem-check-flatness-at-associated-points (source line 7559), invoked at proof lines 7772-7773. Supplies the reduction of flatness of F_T at the generalisation xi to flatness at the associated point x, which closes the contradiction.
- Etale morphisms do not increase the dimension of a subvariety. Descent, Lemma labelled descent-lemma-dimension-at-point-local (Stacks tag 04N4), invoked at proof line 7749. Supplies dim(Z') <= dim(Z) for the etale map Z' -> Z.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original definition 05MT: sources/stacks-flat/pinned-original-source.tex, source lines 6311–6311. Same-chapter pointer into the author environment situation at source line 6311; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 05MU: sources/stacks-flat/pinned-original-source.tex, source lines 6325–6325. Same-chapter pointer into the author environment equation at source line 6325; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02JZ: sources/stacks-chow/tags.txt, source lines 3354–3354. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7746 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 04N4: sources/stacks-chow/tags.txt, source lines 6065–6065. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7749 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 05DC: sources/stacks-chow/tags.txt, source lines 6950–6950. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7763 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original remark 05KL: sources/stacks-chow/tags.txt, source lines 7205–7205. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 7764 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original theorem 05UA: sources/stacks-flat/pinned-original-source.tex, source lines 7559–7559. Same-chapter pointer into the author environment theorem at source line 7559; invoked at proof line 7773 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 05MT. Let $h : X' \to X$ be an étale morphism. Set $\mathcal{F}' = h^*\mathcal{F}$ and $f' = f \circ h$. Let F'_n be (05MU) associated to $(f' : X' \to S, \mathcal{F}')$. Then F_n is a subfunctor of F'_n and if $h(X') \supset Ass_{X/S}(\mathcal{F})$, then $F_n = F'_n$.*

Proof. Let $T \to S$ be any morphism. Then $h_T : X'_T \to X_T$ is étale as a base change of the étale morphism g. For $t \in T$ denote $Z \subset X_t$ the set of points where $\mathcal{F}_T$ is not flat over T, and similarly denote $Z' \subset X'_t$ the set of points where $\mathcal{F}'_T$ is not flat over T. As $\mathcal{F}'_T = h_T^*\mathcal{F}_T$ we see that $Z' = h_t^{-1}(Z)$, see Morphisms, Lemma 02JZ. Hence $Z' \to Z$ is an étale morphism, so $\dim(Z') \leq \dim(Z)$ (for example by Descent, Lemma 04N4 or just because an étale morphism is smooth of relative dimension 0). This implies that $F_n \subset F'_n$.

Finally, suppose that $h(X') \supset \mathrm{Ass}_{X/S}(\mathcal{F})$ and that $T \to S$ is a morphism such that $F'_n(T)$ is nonempty, i.e., such that $\mathcal{F}'_T$ is flat in dimensions $\geq n$ over T. Pick a point $t \in T$ and let $Z \subset X_t$ and $Z' \subset X'_t$ be as above. To get a contradiction assume that $\dim(Z) \geq n$. Pick a generic point $\xi \in Z$ corresponding to a component of dimension $\geq n$. Let $x \in \mathrm{Ass}_{X_t}(\mathcal{F}_t)$ be a generalization of ξ. Then x maps to

a point of $\mathrm{Ass}_{X/S}(\mathcal{F})$ by Divisors, Lemma 05DC and Remark 05KL. Thus we see that x is in the image of h_T, say $x = h_T(x')$ for some $x' \in X'_T$. But $x' \notin Z'$ as $x \rightsquigarrow \xi$ and $\dim(Z') < n$. Hence $\mathcal{F}'_T$ is flat over T at x' which implies that $\mathcal{F}_T$ is flat at x over T (by Morphisms, Lemma 02JZ). Since this holds for every such x we conclude that $\mathcal{F}_T$ is flat over T at ξ by Theorem 05UA which is the desired contradiction. $\square$